SIMCRC CHILE

Colonoscopy capacity approximation for Chile: cohort analysis

How we weighted the simulated results to Chile’s population
Example: FIT every 2 years, ages 45–70

The problem: the simulated population is much bigger than Chile’s

The model simulates one group of about 1 million 40-year-olds until age 100. Chile has a different number of people at every age.

Simulated, alive at 40

971,451

Chile (FONASA), age 40

248,703

Simulated group is bigger by

about 4×

STEP 1

Count Chile’s people at each single age

FONASA gives 5-year groups. We used a linear interpolation of group averages, followed by rescaling within groups.

FONASA group,
ages 40–44

1,171,980

Age	People
40	248,703
41	240,701
42	232,700
43	227,525
44	222,350

STEP 1, WORKED EXAMPLE

How the interpolation works: ages 70–74

1 · A dot at each middle age

Each group's total ÷ 5, placed at the group's middle age.

Group	Per year	Middle
65–69	175,588	67
70–74	**131,771**	**72**
75–79	96,306	77

2 · Read off the line

Straight lines join the dots. Age 70 is 3/5 of the way from 67 to 72.

Age	On the line
70	149,298
71	140,535
72	131,771
73	124,678
74	117,585
Total	**663,868**

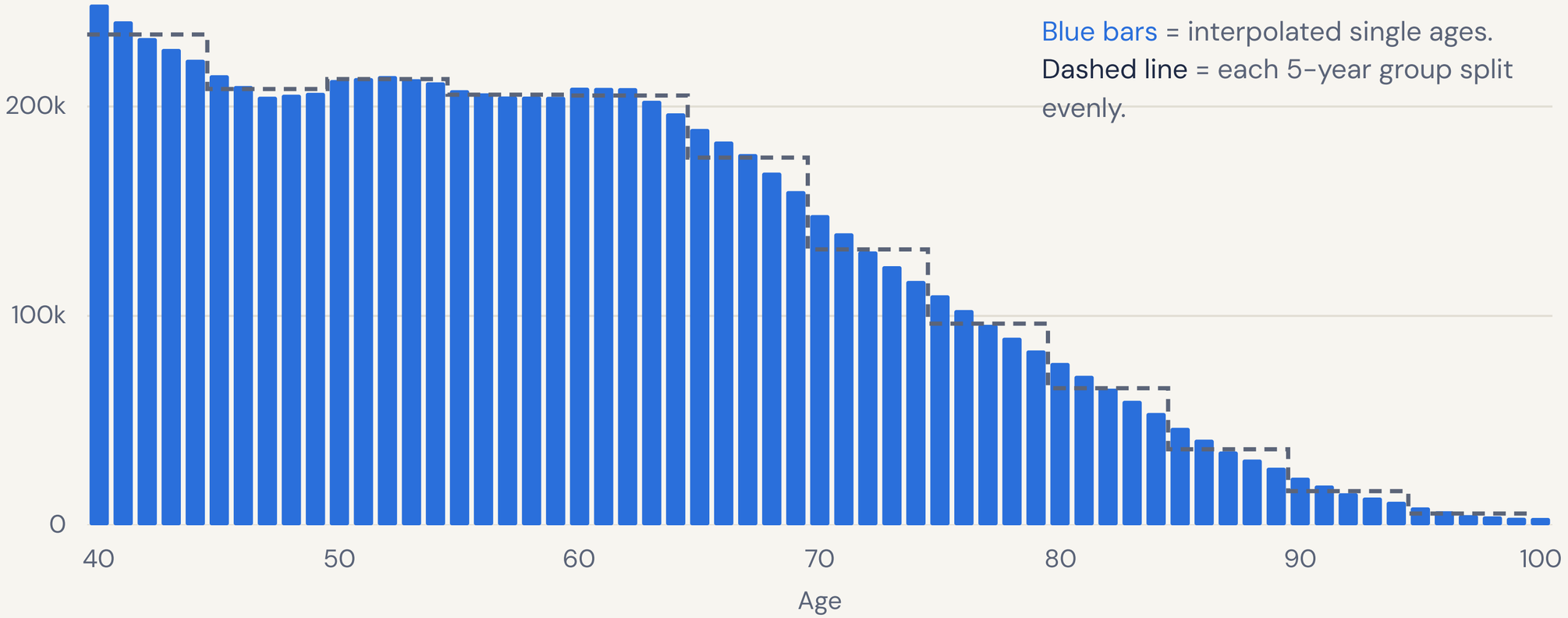
3 · Rescale to the total

Each age × 658,857 ÷ 663,868 = 0.992, so the group matches FONASA.

Age	Final
70	**148,171**
71	**139,474**
72	**130,777**
73	**123,737**
74	**116,698**
Total	**658,857 ✓**

STEP 1, THE RESULT

Chile’s population at each age, 40–100



STEP 2

Divide Chile by the simulated people

Weight = Chile's people at that age ÷ simulated people alive at that age (No Screening)

Age	Chile (FONASA)	Simulated, No Screening	Weight
40	248,703	971,451	0.26
55	207,782	933,083	0.22
70	148,171	806,992	0.18
85	46,575	443,379	0.11

STEP 3

Multiply every number by that age’s weight

Example: FIT tests with a FIT every 2 years, ages 45–70. The same is done for colonoscopies, cancers, deaths, life-years and every other column.

Age	Simulated FIT tests	× Weight	= Chile FIT tests
45	962,809	× 0.223	214,870
55	755,670	× 0.223	168,275
65	641,044	× 0.218	139,950
69	581,266	× 0.194	113,018

WHAT WE COUNT

Year one of a theoretical program

Picture one year where everyone at a screening age (45, 47, … 69) does their FIT, and everyone with a positive FIT gets a colonoscopy.

Counted

Follow-up colonoscopies after a positive FIT, at the exact screening ages.

In a colonoscopy program, this would be the screening colonoscopies.

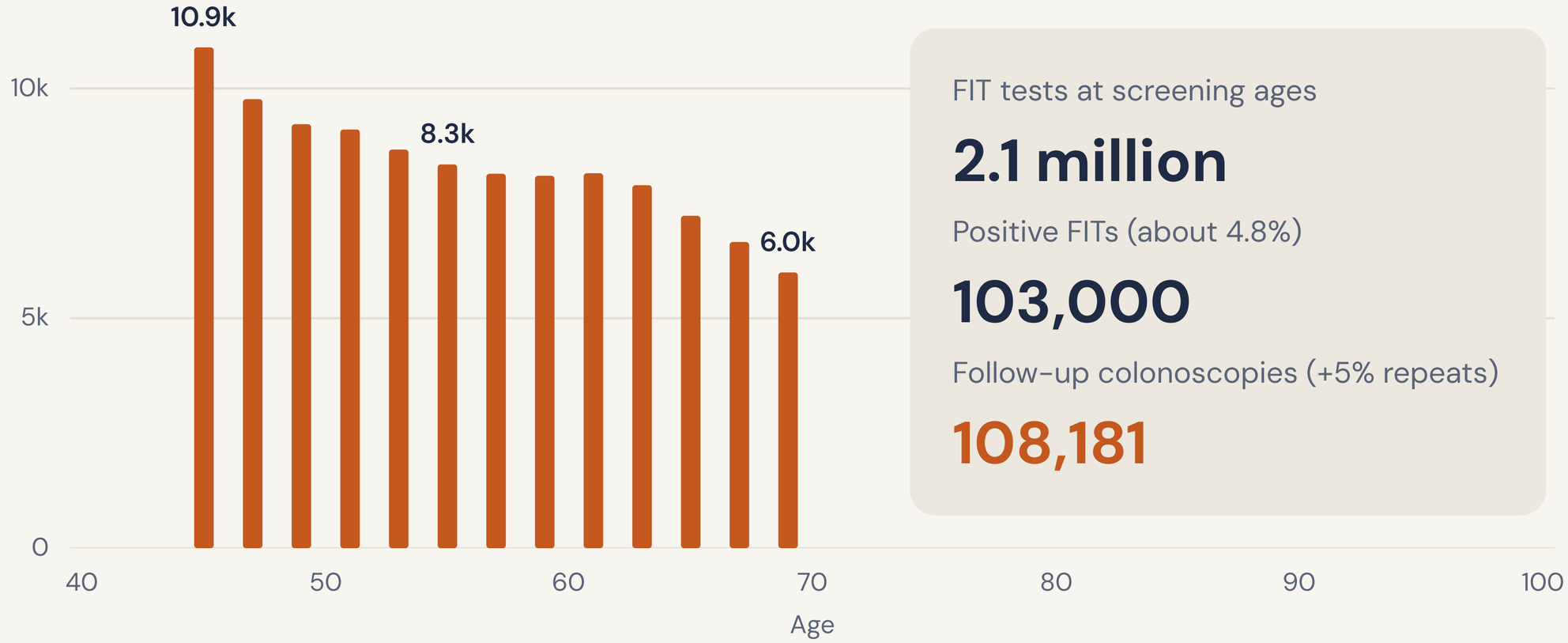
Not counted

Surveillance colonoscopies and colonoscopies for symptoms.

Surveillance only builds up after a program has run for years; symptom colonoscopies happen with or without a program.

THE RESULT

108,000 colonoscopies in year one



FIT tests at screening ages

2.1 million

Positive FITs (about 4.8%)

103,000

Follow-up colonoscopies (+5% repeats)

108,181

CAPACITY

No screening vs FIT every 2 years

No screening

0

FIT every 2 years, 45–70

108,181

Extra colonoscopies in year one

+108,181

Ages	FIT tests	Follow-up colonoscopies
45–49	596,744	29,881
50–59	851,476	42,364
60–69	694,995	35,936
All	2,143,216	108,181

KEEP IN MIND

108,181 is a rough approximation

No gradual rollout

Estimate too high

The Netherlands phased in its national program over 5–6 years. FIT programs front-load colonoscopies; a negative one protects for about 10 years.

100% uptake and adherence

Estimate far too high

Everyone does every FIT and every follow-up. Real uptake is much lower, so this widely overestimates colonoscopies.

Surveillance builds up

Estimate too low, in later years

No surveillance in a true year one, but polyps found now bring surveillance colonoscopies in every later year.

Off-schedule ages

Estimate too low

We count only exact screening ages (45, 47, …). People aged 46, 48, … would also become eligible and likely join early.

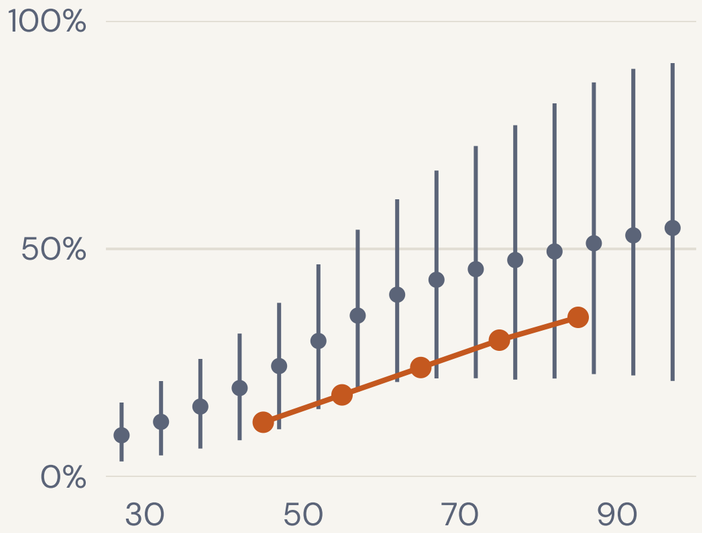
NEXT STEP: A POPULATION MODEL

Does the Mexico City model fit Chile?

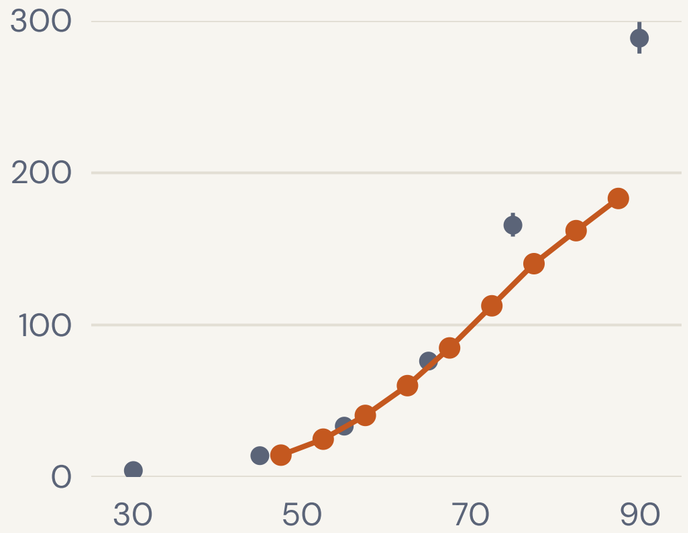
Mexico City model
Chile target and range

Stage: three dots per bar are proximal colon, distal colon and rectum

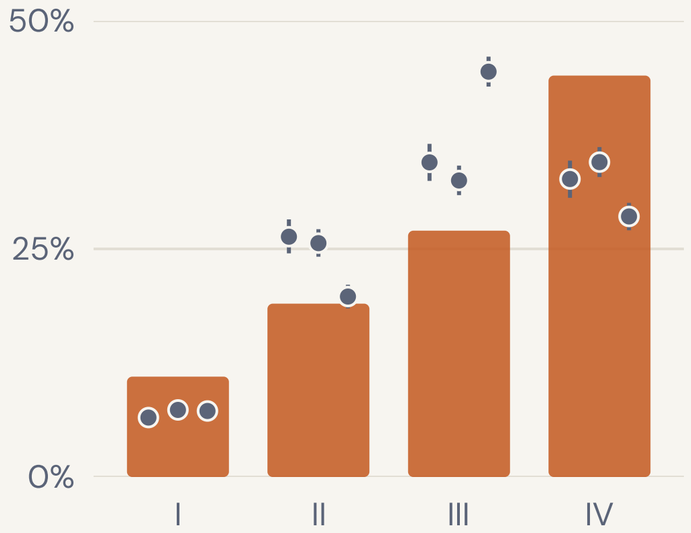
Adenoma prevalence



Cancer incidence



Stage at diagnosis



Good fit to age 69

Incidence is close at ages 40–69, the ages that matter most for FIT 45–70.

Too low from age 70

About 126 vs 166 per 100,000 at 70–79, and 173 vs 289 at 80+.

Later stage

44% of cancers at stage IV, vs about 29–35% in Chile.

POPULATION MODEL

Assumptions in the Mexico City model

1

Realistic, delayed uptake and adherence

Not everyone does their FIT, and many who do it are late. Based on a FIT pilot study in Mexico City with education and in-person home visits, so still optimistic.

Replaces 100%, on-time uptake

2

Program phased in over 5 years

Eligible people are brought into the program gradually, rather than all in year one.

Replaces no rollout

3

Everyone aged 45–70 can start

All eligible people can join, not only those at the exact screening ages (45, 47, …).

Replaces off-schedule ages

COHORT VS POPULATION MODEL

About 6× fewer colonoscopies in year one

Cohort approximation

108,181

Year one, 100% uptake, no rollout, exact screening ages only

Mexico City population model, Chile's population

18,394

Year one, pilot-based uptake, 5-year phase-in, everyone aged 45–70

Difference

5.9×

The cohort estimate is 5.9× year one, and 3.3× the year 5 peak.

Population model, colonoscopies by program year: demand rises to its maximum in year 5, the capacity to plan for

Year 1 · 2025	Year 2 · 2026	Year 3 · 2027	Year 4 · 2028	Year 5 · 2029 · peak
18,394	23,550	26,882	30,884	33,016

LIMITATIONS

What a cohort run misses

Issue	Cohort run (this approximation)	Full population run
Program years	One number, no calendar time	Demand each year: 18,394 → 33,016 by year 5
Rollout	Everyone starts at once	Phased in over 5 years
Uptake	100%, always on time	Realistic and delayed, from the CDMX pilot
Who can start	Only exact screening ages (45, 47, …)	Everyone aged 45–70
Screening history	Older ages already screened since 45	Everyone unscreened in year one
Birth cohorts	One cohort stands in for every age	Each birth cohort, with its own size and mortality
Population	FONASA 2025, held fixed	Chile's projections, growing and ageing

Net result: the cohort run overstates year-one demand about 6×.

POPULATION MODEL: UPTAKE AND ADHERENCE

Out of 100 eligible people

30
Never screen

70
Ever screen. Not all at once: uptake builds up gradually to 70% over the first 5 years

Zoom in: of the 70 who ever screen

50%
Only one FIT

25%
Every other FIT

25%
Every FIT, on schedule

65%
get a follow-up colonoscopy after a positive FIT

65%
go to surveillance colonoscopies, the same as follow-up

Sources: Mexico City FIT pilot study (about 70% returned their FIT; 65% follow-up colonoscopy). Repeat FIT patterns from a US study, since no Mexican data exist: about 24% on schedule, 47% only one FIT.

POPULATION MODEL:
ROLLOUT

Oldest birth cohorts start first

Each cell is a birth cohort's age in that program year. 6 new cohorts join each year, oldest first, and stay eligible until age 70.

Grey = eligible for screening that year

Cohorts eligible: 6 in 2025, rising to all 26 in 2029.

	Program year				
Born	2025	2026	2027	2028	2029
1984					45
1983					46
1982					47
1981					48
1980					49
1979					50
1978				50	51
1977				51	52
1976				52	53
1975				53	54
1974				54	55
1973				55	56
1972			55	56	57
1971			56	57	58
1970			57	58	59
1969			58	59	60
1968			59	60	61
1967			60	61	62
1966		60	61	62	63
1965		61	62	63	64
1964		62	63	64	65
1963		63	64	65	66
1962		64	65	66	67
1961		65	66	67	68
1960	65	66	67	68	69
1959	66	67	68	69	70
1958	67	68	69	70	
1957	68	69	70		
1956	69	70			
1955	70				